Vitamin D for Improved Bone Health and Prevention of Stress Fractures: A Review of the Literature

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Abstract
Vitamin D is a vital nutrient and hormone needed for many essential functions in overall health. There is growing literature examining the role of vitamin D not only in the general population but also in athletes. The most predominantly studied area of vitamin D pertains to bone health. Recently, there has been increased investigation into the relationship of vitamin D and stress fractures, including genetic polymorphisms, levels of 25-hydroxyvitamin D, and bioavailable vitamin D. This review will address the most recent developments of vitamin D research and its important role in bone health in athletes.

Introduction

Stress fractures and overall bone health are a significant topic of interest in the active population, especially endurance athletes and military recruits, because of potential participation time lost and subsequent impact on performance. There are many factors that contribute to bone health, including genetics, nutrition, type of exercise, smoking, alcohol use, medications, calcium status, and vitamin D status. Vitamin D is well studied and has numerous important roles in the body. It has been found to influence immune function (1), muscle function and injury (2), athletic performance (3), and most notably, bone health. In this review, we will first address vitamin D and its involvement in bone health and then focus on its role in stress fractures, specifically in adolescents and active individuals.

Physiology

Vitamin D functions via both the endocrine and autocrine systems (4). Its function in the endocrine system regulates calcium and phosphorus levels in the blood and contributes to calcium homeostasis. The active form, calcitriol (1,25(OH)D3), increases absorption of calcium and phosphorus from the kidneys and small intestine. In the setting of deficient serum calcium, vitamin D and parathyroid hormone (PTH) will increase intestinal absorption and increase bone turnover to raise calcium levels through osteoclastogenesis (5). It has been found that vitamin D is needed for normal bone trabeculae, as well as callus formation for bone reparation (4). Evidence suggests that 1,25(OH)D3 may play an active role in modulating expression of mineralization factors and can prompt osteocytes to control calcium and phosphate release from the bone (6). Vitamin D has an inverse relationship with PTH, which is responsible for bone resorption and increasing serum calcium levels. Some guidelines use the optimal 25(OH)D serum concentration as the level that maximally suppresses PTH (7). Vitamin D levels depend on not only production via UVB exposure and dietary intake but also levels of pollution (8) and increasing age (5).

Vitamin D can be obtained via dietary sources in the event it is not produced sufficiently with skin exposure to UVB radiation. It is estimated that 5 to 30 min of sun exposure to arms and legs will produce adequate vitamin D, but it is dependent on skin pigmentation, season, latitude, and time of day (5). Nevertheless, exposure to the sun and UVB radiation does have risks, especially in excess amounts. Vitamin D can be found in natural products, such as fish, eggs, and mushrooms (4). In addition, more regularly consumed foods are fortified with vitamin D. For example, in the United States, the largest dietary source of vitamin D is cereal (9). Dairy is an excellent source of vitamin D as well. Due to the intimate and direct relationship with calcium, it can be difficult to exclusively assess the connection of vitamin D with bone health. However, it has been demonstrated that dietary vitamin D, and not dietary calcium, is related to a lower risk of incident stress fracture among girls (10). Further, Americans have among the highest calcium intake in the world but also one of the highest rates of osteoporosis (11).

Magnesium also has a confirmed relationship with vitamin D. Magnesium supplementation has been shown to

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reduce resistance to vitamin D treatment (12). In addition, when magnesium homeostasis is not maintained, the effectiveness and clinical benefits of vitamin D can be significantly reduced (12).

Measured Serum Values

The mostly widely accepted metric for assessing vitamin D status is with serum 25(OH)D. It is a superior method compared with measuring the bioactive form 1,25(OH)D because of a longer half-life, with 21 d to 30 d for 25(OH)D (13) versus 4 h to 6 h (14). In addition, while it is not an active metabolite, it is a total of both D2 and D3 precursors and, therefore, a better representation of intake/production (13). Numerous organizations use this marker for classification of vitamin D sufficiency (Table 1). These recommendations are literature-derived estimations for bone health maintenance, including the relationship of PTH and 25(OH)D levels, efficiency of calcium absorption from the small intestine, and reducing the risk of fracture. The studies used for these recommendations, however, do not take a specific focus on athletes.

Evidence is inconsistent regarding whether athletes have same prevalence of vitamin D insufficiency and deficiency as the general population. This may be due to the numerous factors associated with vitamin D status, which includes nutrition and supplementation, genetics, and most importantly, sunlight exposure. As summarized by Ogan and Pritchett (15), the main risk factor for vitamin D deficiency is being an indoor athlete and the amount of sunlight exposure. Multiple studies show that indoor athletes are at greater risk for insufficiency and deficiency (15). Fishman et al. (16) found 79.3% of the National Basketball Association (NBA) combine participants from 2009 — 13 vitamin D-deficient or insufficient. Bauer and colleagues (17) evaluated elite handball athletes in July and established 31 of 70 had insufficient levels. However, the literature does not just describe indoor athletes with relatively high rates of insufficiency or deficiency. Greater than one third of NCAA Division 1 athletes from multiple sports had 25(OH)D levels <32 $ng \cdot mL^{-1}$ in the summer months (18). National Football League (NFL) players have demonstrated high levels of insufficiency (19), especially at spring practice (20). Further, a systematic review found that 56% of 2313 athletes in multiple nations had vitamin D inadequacy (21). In contrast, a cross-sectional study by Wentz et al. (22) including 59 female distance runners in southeast United States observed adequate vitamin D status with only 18.6% insufficient or deficient (<75 $nmol \cdot L^{-1}$).

Vitamin D and Bone Health

A prominent area of interest is the relationship of vitamin D and bone health. There are multiple measures of bone health including fractures, bone mineral density (BMD), and bone mineral content (BMC). BMD ($mg \cdot cm^2$) and BMC ($mg \cdot mm^{-1}$) are typically calculated with dual-energy X-ray (DEXA), peripheral quantitative computed tomography (pQCT), or high-resolution pQCT (HR-pQCT). While BMD is just one measure of bone health and not necessarily equivalent to fracture risk, it was found that increasing the BMD by 5.4% is equal to a 64% increase in ultimate force and a 94% increase in energy to failure (23). Multiple studies have shown a correlation with BMD, BMC, and 25(OH)D (24–27).

Most of the higher-quality evidence on vitamin D status and bone health outcomes pertain to studies of postmenopausal women and men older than 60 years. In 2005, the Office of Dietary Supplements of the National Institutes of Health and the Agency for Healthcare Research and Quality funded a systematic review of the evidence on the efficacy and safety of vitamin D in relation to bone health outcomes (28). The general trend was fair or inconsistent evidence of an association between 25(OH)D concentration and baseline or change in BMD or BMC (28). There were 17 randomized controlled trials evaluating the effects of vitamin D supplementation on BMD, concluding that at low-dose supplementations of 300 IU or 400 IU daily, there was no significant effect on BMD, while at higher-dose supplementation of more than 700 IU daily, there was consistent evidence that supplementation prevented bone loss. The combined results from 13 individually randomized trials (n = 58,712), evaluating the effect of vitamin D supplementation on fractures, resulted in a nonsignificant reduction in fractures with heterogeneity of treatment effect.

In a randomized controlled trial, Doetsch et al. (29) followed callus formation by means of BMD in osteoporotic proximal humeral fractures and showed taking 800 IU vitamin D3 and 1000 mg calcium was associated with significantly increased BMD and callus formation vs placebo. A meta-analysis of randomized controlled trials (RCTs) investigating vitamin D supplementation found a reduction in risk

Table 1.
Vitamin D status guidelines.

Organization	Sufficiency Level	25(OH)D ($ng \cdot mL^{-1}$)
Vitamin D Council	Deficiency	0 to 30
	Insufficiency	31 to 39
	Sufficiency	40 to 80
	Toxicity	>150
Endocrine Society	Deficiency	0 to 20
	Insufficiency	21 to 29
	Sufficiency	30 to 100
Institute of Medicine Food and Nutrition Board	Deficiency	0 to 11
	Insufficiency	12 to 20
	Sufficiency	>20

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of osteoporotic hip and nonvertebral fractures with a dose of 700 IU·d^{-1} to 800 IU·d^{-1} (30). However, in a 2014 systematic review that evaluated vitamin D in human fracture healing, it was determined that the available data are too inconsistent, and therefore, the clinical role is inconclusive, especially considering most clinical studies use combination treatment (31).

A recent meta-analysis of 81 RCTs (n = 53,537) by Bolland et al. (32) assessed the effects of vitamin D on fractures (42 studies), falls (37 studies), and BMD (41 studies). The conclusion reached was that vitamin D supplementation does not prevent fractures or falls nor does it have clinically meaningful effects on BMD. Therefore, it may detract from more cost-effective treatments if universal supplementation is used in older community dwelling adults. However, there are some criticisms of the article, including a majority of the RCTs with a duration of 1 year or less and most participants consisting of community-dwelling women 65 years or older. Considering there were no individual participant data available and with the subgroup assignments based on the mean baseline 25(OH)D concentration of the given study population, there is a potential for misclassification bias as well.

Vitamin D and Bone Health in Adolescents and Athletes

Adolescent-aged athletes, particularly females, are an important population to study as it relates to vitamin D and bone health. Approximately 90% of peak bone mass is obtained by age 18 years [Whiting et al. (33); Matkovic et al. (34)]. Baxter-Jones et al. (35) have demonstrated peak BMC at the femoral neck is reached in girls at age 15 and 16.5 years in boys. Up to 60% of risk of osteoporosis can be related to bone mineral a by early adulthood, and gains made during the development period prior to peak bone mass can have a large effect on prevention of osteoporotic fractures (34). Therefore, optimizing conditions for mineralization and bone growth in adolescents is ideal for prevention of future bone-related health conditions.

In a randomized, double-blind, placebo-controlled trial, Ghazal and colleagues (27) studied 167 adolescents, 86 girls and 81 boys with a mean age of 13 years, who received vitamin D or placebo. Girls who had received vitamin D had significantly larger hip BMC increments compared with those assigned to placebo, at 24 months compared with study entry, but not at 24 months compared with 12 months. There were no significant differences in bone mass changes between treatment groups in boys, at 24 months compared with 12 months or to baseline.

In a 3-year prospective study, Lehtonen-Veromaa and colleagues (36) examined the association between changes in BMD and serum 25(OH)D in 171 healthy Finnish girls aged 9 years to 15 years. Results showed that baseline 25(OH)D correlated significantly with the unadjusted 3-year change in BMD at the lumbar spine and femoral neck, concluding that pubertal girls with hypovitaminosis D seem to be at risk of not reaching maximum peak bone mass. Furthermore, results from Ward and colleagues (37) showed that there were no effects of vitamin D supplementation on bone in postmenarchal females. The authors conducted a community-based, double-blind, randomized controlled trial of postmenarchal 12- to 14-year-old girls, determining the effect of vitamin D supplementation on bone, as measured using DEXA and pQCT. The authors concluded that there is a lack of effect of intervention after the period of peak mineral accretion, suggesting that earlier action is required. El-Hajj Fuleihan and colleagues (26), in a double-blind, placebo-controlled 1-year trial drew similar conclusions to Ward and colleagues. The authors randomized girls, ages 10 years to 17 years, to receive weekly vitamin D at doses of 1400 or 14,000 IU. In premenarcheal girls, there was a consistent trend for increments in BMD and/or BMC at several skeletal sites, reaching significance at the lumbar spine BMD in the low-dose group, and at the trochanter BMC in both treatment groups. There was no change in BMD or BMC in postmenarcheal girls.

In addition to BMD, HR-pQCT can be used to compute volumetric BMD (vBMD). It also can allow for analysis of cortical and trabecular properties separately as well as assessment of trabecular bone microarchitecture. Cheung et al. (38) used HR-pQCT to correlate 25(OH)D status with multiple bone parameters, including areal BMD and BMC of bilateral femoral necks, cortical area, cortical thickness, trabecular area, and trabecular thickness. The study involved 563 healthy girls and boys aged 12 years to 16 years. The authors noted a significant correlation between 25(OH)D levels and areal BMD, BMC, total vBMD, cortical area, cortical thickness, and trabecular thickness in the girls. In the boys, there was a significant correlation between 25(OH)D levels and areal BMD, BMC, total vBMD, cortical area, cortical thickness, trabecular vBMD, and bone volume-to-tissue volume ratio.

A cohort study of NFL players found that vitamin D levels were significantly lower in players with at least one fracture compared with none when controlling for number of professional years played (19). Silk and colleagues (39), in a 2015 randomized, double-blind, placebo-controlled trial, used pQCT to establish whether vitamin D supplementation would improve bone properties of young male jockeys. Results showed that the supplemented group displayed greater postintervention bone properties, including greater cortical content, larger cortical area, greater cortical density, and greater total area, indicating beneficial effects of supplementation on bone properties in as short as 6 months.

While 25(OH)D is the established marker of vitamin D status, there has been recent evidence that suggests it may not be the most accurate measure to determine bone health (40). Calculating the bioavailable vitamin D from serum 25(OH)D and vitamin D-binding protein (DBP) appears to correlate better with BMD in a diverse population of 604 athletes in Qatar (41). Tenforde et al. (42) recently performed a cohort study on 28 male athletes and likewise did not find an association with 25(OH)D insufficiency and BMD.

Bone Stress Injuries and Stress Fractures

Bone stress injuries (BSIs) are overuse injuries that occur when skeleton fails to withstand submaximal forces acting over time. With day-to-day physiologic stress, bone undergoes normal and accelerated remolding, conditions that are adaptive and clinically asymptomatic. However, as the rate of loading exceeds the rate of remodeling, microdamage can accumulate. Pathology occurs on a continuum of injury, progressing through the stages of stress injury, stress reaction, and stress fracture. Stress fracture represents the final step along the continuum and is defined as a discontinuity in the bony cortex.

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BSIs account for up to 20% of all injuries treated in sports medicine clinics (43) and are associated with pain, reduced performance, lost training time, and medical expense (44). One study found that almost one third of competitive runners have a history of prior stress fracture (45), and in competitive track and field athletes, a 12-month prospective study found a stress fracture rate of 21.1% (46).

Risk factors for BSIs include both extrinsic and intrinsic factors. Extrinsic factors include training load and pattern, training surface, type of sport, and footwear. BSIs are more common in endurance athletes especially after an abrupt increase in frequency, duration, or intensity of activity. Intrinsic factors include anthropometrics, such as age, sex, weight, and height, as well as muscle strength, neuromuscular recruitment, skeletal morphology, and activity of osteoblasts and osteoclasts. Likewise, the level of vitamin D is a product of multiple extrinsic and intrinsic factors, as previously mentioned.

Chatzipapas et al. (47) published a novel case control study in 2009, comparing the frequency of certain vitamin D receptor (VDR) genes in 32 male military personnel with lower-extremity stress fractures with 32 matched healthy controls. The stress fractures were diagnosed with plain radiographs or a bone scan. It was found that an allele of FokI was significantly more frequent in patients with stress fractures than controls, and an allele of BsmI showed a tendency without reaching statistical significance. This study identified a potential genetic component to stress fractures with relation to vitamin D. It did not, however, assess 25(OH)D levels nor address vitamin D supplement status in the study participants.

In a 2016 retrospective cohort study, Miller and colleagues (48) demonstrated an association between patients, mean age of 44 years and 66% female, with stress fractures and serum vitamin D concentration less than 40 $ng{\cdot}mL^{-1}$. Smith and colleagues (49) demonstrated that 31 patients with stress fractures with a mean age of 52 years had a vitamin D level that was significantly lower compared with 41 patients with ankle sprains and no fracture. Both the Miller and Smith studies did not specifically investigate athletes.

While there is an overall paucity of placebo-controlled RCTs that specifically address the athletic population as it pertains to the association of vitamin D levels and stress fractures, there is an overall consistency suggesting an association between vitamin D concentration and BMC and stress fractures.

Association between Vitamin D Status/Supplementation and Bone Health/Stress Fracture Outcomes in the Active Population (Military and Athletics)

An active population in which there is research investigating the relationship of vitamin D and bone health involves military recruits. Ruohola et al. (50) followed 756 randomly selected Finnish military recruits for more than 90 d and assessed for the relationship of baseline 25(OH)D concentration and risk of developing a stress fracture (50). Blood samples were taken in early July, and no vitamin D supplementation was used by any participants. Twenty-two stress fractures were diagnosed with radiographical evidence during the 90 d. The median 25(OH)D level in those that sustained a stress fracture was significantly lower than those without stress fracture (64.3 $nmol{\cdot}L^{-1}$ vs 76.2 $nmol{\cdot}L^{-1}$, $P = 0.017$). The overall median serum 25(OH)D level was 75.8 $nmol{\cdot}L^{-1}$, subjects lower than the median had a significantly greater risk for stress fracture ($P = 0.002$) than those above the median.

A systematic review and meta-analysis, published by Dao and colleagues (51), examined the association between serum 25(OH)D levels and stress fractures specifically in the military. The analysis included 8 studies published between 2000 and 2013 consisting of 2634 military personnel, with 761 stress fracture cases and 1873 controls. The authors found that the overall mean serum 25(OH)D level was significantly lower for stress fracture cases than controls, with a pooled mean difference of −2.44 $ng{\cdot}mL^{-1}$ (95% confidence interval, −4.05 to −0.84) (48). The 25(OH)D levels also were significantly lower in stress fracture cases than controls at the time of diagnosis (Fig. 1). Other studies investigating the military population have demonstrated similar findings to the meta-analysis. Davey and colleagues (52), in a 2015 prospective study following 1082 Royal Marine recruits, concluded that baseline serum 25(OH)D concentration below 50 $nmol{\cdot}L^{-1}$ was associated with an increased risk of stress fractures. A study involving 37 British Army recruits demonstrated a shorter recovery time from lower-limb stress fractures with sufficient 25(OH)D at the time of injury, with a statistically significant mean difference of 17.8% in recovery time between recruits with sufficient 25(OH)D (>50 $nmol{\cdot}L^{-1}$) and insufficiency (<50 $nmol{\cdot}L^{-1}$) ($P = 0.034$) (53). The mean recovery time in weeks was 10.1 (±2.5) in the 25(OH)D sufficient group, 11.9(±2.4) in the insufficient group (25 to 50 $nmol{\cdot}L^{-1}$), and 13 (±1.6) in the deficient group (<25 $nmol{\cdot}L^{-1}$).

In a cumulative case-control study, Shimasaki and colleagues (54) evaluated serum 25(OH)D concentration in 37 Japanese male university soccer athletes, 18 of which had a fifth metatarsal stress fracture and 19 controls. Results showed that 25(OH)D levels less than 30 $ng{\cdot}mL^{-1}$ were associated with statistically significantly increased odds of stress fracture.

In a study in which the authors evaluated young female athletes ages 18 years to 26 years, Nieves and colleagues (55) followed subjects for an average of 1.85 years and measured longitudinal changes in bone density and incident stress fractures. The authors found that vitamin D intake predicted gains in spine and hip BMD. Additionally, the study demonstrated the protective effects of vitamin D on stress fracture risk in female athletes, with an adjusted hazard ratio of 0.67 (0.34 to 1.31). The authors discussed that it is not surprising that many of the same nutrients, vitamin D included, which were related to increases in BMD, were found to be protective against stress fractures.

Findings from Nieves et al. are further supported by a report in female Navy recruits. In this interventional trial, women were randomized to supplementation with 2000 mg calcium and 800 international units of vitamin D versus placebo for 8 wk of basic training (56). Those who had supplementation had a 20% lower incidence of stress fracture.

Supplementation

It is not known if an optimal 25(OH)D level for athletes differs from that of the general population. In general, secondary to the inaccuracy and imprecision of the different assays and the lack of a validated method for measuring vitamin D levels, generalizable conclusions are difficult to make. Shuler et al. (57) recommends supplementation in the athlete with 25 (OH)D below 30 $ng{\cdot}mL^{-1}$. The Female Athlete Triad

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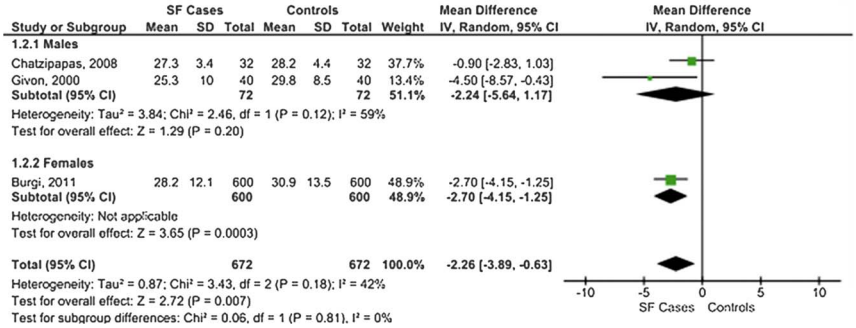

Figure: Comparison of serum 25(OH)D levels measured between SF cases and controls at diagnosis. Dao et al. (48). SF, stress fracture.

Coalition recommends maintaining levels between 32 ng·mL^{-1} and 50 ng·mL^{-1} (58). Further, Cannell and colleagues (3) suggest that, based on improvements in athletic performance, the ideal level for 25(OH)D in athletes may be as high as 50 ng·mL^{-1}. In terms of supplementation dosage, McCabe et al. (59) recommend at least 800 to 2000 IU daily if at high risk of stress fracture, especially during winter or spring months.

While absolute conclusions are difficult to make from a rather incongruous set of studies, there is an overall trend showing an association between vitamin D supplementation and bone health outcomes. In addition, vitamin D3 has been shown to be more effective than D2 in maintaining 25(OH)D status (5,60,61). Therefore, we recommend vitamin D3 supplementation to maintain 25(OH)D status, especially in athletes at high risk of stress fracture. As it relates to bone health in the athletic population, the data are not strong enough to support a dose or target serum level, although, in general, the studies support higher doses to promote bone health.

There are sufficient data to support the safety profile of vitamin D supplementation, even in doses above commonly recommended ranges. Cranney et al.'s (28) 2008 summary reported on 22 trials that assessed adverse events associated with vitamin D supplementation. Biochemical abnormalities, such as hypercalcemia and hypercalciuria, were the most frequently reported, although the rates of these events between vitamin D and placebo were not significant and not clinically relevant. Of seven trials that reported kidney stone incidence, one reported an absolute increase in kidney stones in women. Overall, there is fair evidence that adults tolerate vitamin D at doses above current dietary reference intake levels (28). Therefore, from a safety perspective, the evidence supports higher doses of vitamin D supplementation.

Table 2.
Conversion table for serum vitamin D.

ng·mL^{-1}	nmol·L^{-1}
10	25
12	30
16	40
20	50
24	60
28	70
30	75
36	90
40	100
50	125

Conversion formula: [nmol·L^{-1}] = 2.5 × [ng·mL^{-1}].

Future Directions

25(OH)D has long been the serum measurement of choice both clinically and in the literature for gauging vitamin D status. Considering recent work investigating VDR polymorphisms, DBP, and calculating bioavailable vitamin D, it is possible that 25(OH)D will no longer be the most optimal measurement. This may be the case especially in athletes who need a truly accurate assessment, such as if they are at a higher risk for stress fractures or have markers of poor bone health. More studies will need to replicate and expand upon the investigation already done examining the relationship of genetics and bioavailable vitamin D regarding bone health and stress fractures in athletes. Considering the role of magnesium in vitamin D activation, it may prove valuable to introduce the evaluation of magnesium status and supplementation into further studies as well.

Conclusion

Literature demonstrates evidence that vitamin D status, bone health, and stress fractures are connected, although studies are inconsistent due to heterogeneity. Evidence is stronger in populations that are not sufficient in vitamin D and are at high risk of stress fracture. Vitamin D has a high therapeutic index with a good safety profile. Vitamin D optimization should be strongly considered especially in athletes that are at high risk of stress fractures. Additionally, higher doses of vitamin D can be considered when clinically indicated. Further research is needed to improve on the current heterogeneity and include more recent developments, including polymorphisms, bioavailable vitamin D, magnesium status, and HR-pQCT.

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